Polarization Interference

Interference of two (monochromatic) optical fields with arbitrary polarization

Wavevector difference

Let the two time-harmonic electric-field vectors impinging on the same detector be

$$\mathbf{E}_1(\mathbf{r},t) = \mathbf{e}_1 A_1 \exp\left[i\left(\mathbf{k}_1 \cdot \mathbf{r} - \omega t + \phi_1\right)\right]$$
$$\mathbf{E}_2(\mathbf{r},t) = \mathbf{e}_2 A_2 \exp\left[i\left(\mathbf{k}_2 \cdot \mathbf{r} - \omega t + \phi_2\right)\right] \tag{1}$$

where A_1 and A_2 are amplitudes, ϕ_1 and ϕ_2 are initial phases, and $\mathbf{e}_1$, $\mathbf{e}_2$ are complex Jones unit vectors that give the polarization state ($\|\mathbf{e}_1\| = 1$, $\|\mathbf{e}_2\| = 1$).

The instantaneous total field is

$$\mathbf{E} = \mathbf{E}_1(\mathbf{r},t) + \mathbf{E}_2(\mathbf{r},t) \tag{2}$$

And the time-average intensity is

$$\boldsymbol{I} = \left\langle |\mathbf{E}|^2 \right\rangle = A_1^2 + A_2^2 + 2A_1A_2\Re\left\{\left(\mathbf{e}_1 \cdot \mathbf{e}_2^*\right)\exp\left[i\left(\mathbf{k}_1 - \mathbf{k}_2\right)\cdot\mathbf{r} + i\left(\phi_1 - \phi_2\right)\right]\right\} \tag{3}$$

And the interference term is

$$\boldsymbol{I} = 2A_1A_2\left|\mathbf{e}_1 \cdot \mathbf{e}_2^*\right|\cos\left[\left(\mathbf{k}_1 - \mathbf{k}_2\right)\cdot\mathbf{r} + \left(\phi_1 - \phi_2\right) + \arg\left(\mathbf{e}_1 \cdot \mathbf{e}_2^*\right)\right] \tag{4}$$

Path-length difference

Let the two time-harmonic electric-field vectors impinging on the same detector be

$$\mathbf{E}_1(\mathbf{r},t) = \mathbf{e}_1 A_1 \exp\left[i\left(\mathbf{k} \cdot \mathbf{r}_1 - \omega t + \phi_1\right)\right]$$
$$\mathbf{E}_2(\mathbf{r},t) = \mathbf{e}_2 A_2 \exp\left[i\left(\mathbf{k} \cdot \mathbf{r}_2 - \omega t + \phi_2\right)\right] \tag{5}$$

where A_1 and A_2 are amplitudes, ϕ_1 and ϕ_2 are initial phases, and $\mathbf{e}_1$, $\mathbf{e}_2$ are complex Jones unit vectors that give the polarization state ($\|\mathbf{e}_1\| = 1$, $\|\mathbf{e}_2\| = 1$).

The instantaneous total field is

$$\mathbf{E} = \mathbf{E}_1(\mathbf{r},t) + \mathbf{E}_2(\mathbf{r},t) \tag{6}$$

And the time-average intensity is

$$\boldsymbol{I} = \left\langle |\mathbf{E}|^2 \right\rangle = A_1^2 + A_2^2 + 2A_1A_2\Re\left\{\left(\mathbf{e}_1 \cdot \mathbf{e}_2^*\right)\exp\left[i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2) + i(\phi_1 - \phi_2)\right]\right\} \tag{7}$$

And the interference term is

$$\boldsymbol{I} = 2A_1A_2\left|\mathbf{e}_1 \cdot \mathbf{e}_2^*\right|\cos\left[\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2) + (\phi_1 - \phi_2) + \arg\left(\mathbf{e}_1 \cdot \mathbf{e}_2^*\right)\right] \tag{8}$$

Special cases

Special cases

Polarisation vectors	Inner product $\mathbf{e}_1 \cdot \mathbf{e}_2^*$	Result
Identical linear (angle θ = 0)	1	$I = A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\phi$ (maximal fringe contrast)
Linear, angle θ between them	$\cos\theta$	$I_{\text{int}} = 2A_1A_2\cos\theta\cos\Delta\phi$
Orthogonal linear (θ = 90°)	0	No interference term → $I = A_1^2 + A_2^2$
Identical circular (same handedness)	1	Same as first row
Opposite circular	0	No interference

Thus, the fringe visibility is scaled by the polarization overlap factor,

$$\gamma = \left|\mathbf{e}_1 \cdot \mathbf{e}_2^*\right| \quad (0 \le \gamma \le 1)$$

Polarization interference considering reflection coefficients

Assuming the vector of the incident electric field can be expressed as

$$\mathbf{E}(\mathbf{r},t) = E_0\mathbf{e}_x \exp\left[i(\mathbf{k}\cdot\mathbf{r} - \omega t)\right] \tag{9}$$

And the two time-harmonic electric-field vectors after the beam splitter are as,

$$\begin{aligned} \mathbf{E}_{\text{obj}}(\mathbf{r},t) &= \mathbf{e}_1 E_0 \exp\left[i(\mathbf{k}\cdot\mathbf{r}_1 - \omega t + \phi_1)\right] \\ \mathbf{E}_{\text{ref}}(\mathbf{r},t) &= \mathbf{e}_2 E_0 \exp\left[i(\mathbf{k}\cdot\mathbf{r}_2 - \omega t + \phi_2)\right] \end{aligned} \tag{10}$$

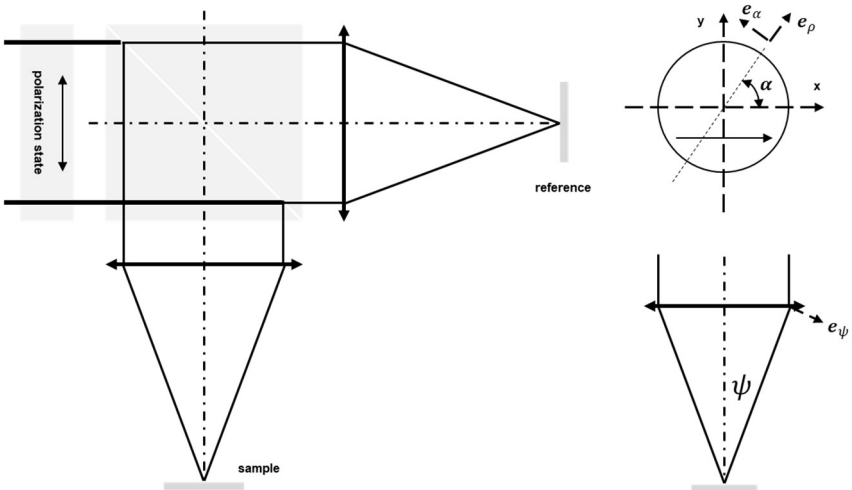
where $\mathbf{e}_1$ and $\mathbf{e}_2$ are the same as,

$$\mathbf{e}_1 = \cos\alpha\,\mathbf{e}_\rho - \sin\alpha\,\mathbf{e}_\alpha \tag{11}$$

And after refraction by the lens, the component along the $\mathbf{e}_\rho$ axis becomes along the $\mathbf{e}_\psi$, that is

$$\mathbf{e}_1 = \cos\alpha\mathbf{e}_\psi - \sin\alpha\mathbf{e}_\alpha$$	**(12)**

On the incident plane, the components of electric field along vectors $\mathbf{e}_\psi$ and $\mathbf{e}_\alpha$ are parallel-polarized and perpendicular-polarized with respect to the reference or sample plane.



There, the reflected vector is expressed as, after reflection by the reference or sample plane,

	$$\mathbf{E}_{\text{obj}}(\psi) = \left(r_{\parallel 1}\cos\alpha\mathbf{e}_\psi - r_{\perp 1}\sin\alpha\mathbf{e}_\alpha\right)E_0\exp\left[i\left(\mathbf{k}_\psi\cdot\mathbf{r}_1+\phi_1\right)\right]$$	**(13)**
	$$\mathbf{E}_{\text{ref}}(\psi) = \left(r_{\parallel 2}\cos\alpha\mathbf{e}_\psi - r_{\perp 2}\sin\alpha\mathbf{e}_\alpha\right)E_0\exp\left[i\left(\mathbf{k}_\psi\cdot\mathbf{r}_2-\omega t+\phi_1\right)\right]$$	

And after refraction by the lens again,

$$\mathbf{E}_{\text{obj}}(\psi) = \left[\left(\cos^2\alpha r_{\parallel 1}+\sin^2\alpha r_{\perp 1}\right)\mathbf{e}_x+\sin\alpha\cos\alpha\left(r_{\parallel 1}-r_{\perp 1}\right)\mathbf{e}_y\right]$$ $$E_0\exp\left[i\left(\mathbf{k}_\psi\cdot\mathbf{r}_1+\phi_1\right)\right] = \left(r_{x1}\mathbf{e}_x+r_{y1}\mathbf{e}_y\right)E_0\exp\left[i\left(\mathbf{k}_\psi\cdot\mathbf{r}_1+\phi_1\right)\right]$$	**(14)**
$$\mathbf{E}_{\text{ref}}(\psi) = \left[\left(\cos^2\alpha r_{\parallel 2}+\sin^2\alpha r_{\perp 2}\right)\mathbf{e}_x+\sin\alpha\cos\alpha\left(r_{\parallel 2}-r_{\perp 2}\right)\mathbf{e}_y\right]$$ $$E_0\exp\left[i\left(\mathbf{k}_\psi\cdot\mathbf{r}_2+\phi_2\right)\right] = \left(r_{x2}\mathbf{e}_x+r_{y2}\mathbf{e}_y\right)E_0\exp\left[i\left(\mathbf{k}_\psi\cdot\mathbf{r}_2+\phi_2\right)\right]$$	

It is seen that the beam is no longer linearly-polarized after reflected from the plane via a lens. Thus, the time-average intensity is

$$
\begin{aligned}
\boldsymbol{I} &= \left\langle \left| \mathbf{E}_{\text{obj}}(\psi) + \mathbf{E}_{\text{ref}}(\psi) \right|^2 \right\rangle \\
&= E_0^2 \left| r_{x1}\mathbf{e}_x + r_{y1}\mathbf{e}_y \right|^2 + E_0^2 \left| r_{x2}\mathbf{e}_x + r_{y2}\mathbf{e}_y \right|^2 + 2E_0^2 \left| \left(r_{x1}\mathbf{e}_x + r_{y1}\mathbf{e}_y\right)\left(r_{x2}\mathbf{e}_x + r_{y2}\mathbf{e}_y\right)^* \right| \\
&\cos\left[\mathbf{k}\cdot\left(\mathbf{r}_1 - \mathbf{r}_2\right) + \Delta\phi_{12} + \arg\left[\left(r_{x1}\mathbf{e}_x + r_{y1}\mathbf{e}_y\right)\left(r_{x2}\mathbf{e}_x + r_{y2}\mathbf{e}_y\right)^* \right] \right]
\end{aligned}
\tag{15}
$$

where

$$
\begin{aligned}
&\left(r_{x1}\mathbf{e}_x + r_{y1}\mathbf{e}_y\right)\bullet\left(r_{x2}\mathbf{e}_x + r_{y2}\mathbf{e}_y\right)^* = r_{x1}\overline{r_{x2}} + r_{y1}\overline{r_{y2}} = \left(\cos^2\alpha r_{\square 1} + \sin^2\alpha r_{\perp 1}\right)\left(\cos^2\alpha\overline{r_{\square 2}} + \sin^2\alpha\overline{r_{\perp 2}}\right) \\
&+\sin\alpha\cos\alpha\left(r_{\square 1} - r_{\perp 1}\right)\sin\alpha\cos\alpha\left(\overline{r_{\square 2}} - \overline{r_{\perp 2}}\right) \\
&= \cos^4\alpha r_{\square 1}\overline{r_{\square 2}} + \sin^4\alpha r_{\perp 1}\overline{r_{\perp 2}} + \cos^2\alpha\sin^2\alpha\left(r_{\square 1}\overline{r_{\perp 2}} + r_{\perp 1}\overline{r_{\square 2}}\right) + \sin^2\alpha\cos^2\alpha\left(r_{\square 1}\overline{r_{\square 2}} + r_{\perp 1}\overline{r_{\perp 2}} - \overline{r_{\square 2}}r_{\perp 1} - r_{\square 1}\overline{r_{\perp 2}}\right) \\
&= \cos^4\alpha r_{\square 1}\overline{r_{\square 2}} + \sin^4\alpha r_{\perp 1}\overline{r_{\perp 2}} + \sin^2\alpha\cos^2\alpha\left(r_{\square 1}\overline{r_{\square 2}} + r_{\perp 1}\overline{r_{\perp 2}}\right) = \cos^2\alpha r_{\square 1}\overline{r_{\square 2}} + \sin^2\alpha r_{\perp 1}\overline{r_{\perp 2}} \\
&= \cos^2\alpha\left|r_{\square 1}r_{\square 2}\right|\mathrm{e}^{i\left[\angle r_{\square 1} - \angle r_{\square 2}\right]} + \sin^2\alpha\left|r_{\perp 1}r_{\perp 2}\right|e^{i\angle r_{\perp 1} - \angle r_{\perp 2}}
\end{aligned}
$$

Thus，Eq. (15) can be expressed as

$$
\begin{aligned}
\boldsymbol{I} &= E_0^2 \left| r_{x1}\mathbf{e}_x + r_{y1}\mathbf{e}_y \right|^2 + E_0^2 \left| r_{x2}\mathbf{e}_x + r_{y2}\mathbf{e}_y \right|^2 + 2E_0^2 \left| \left(r_{x1}\mathbf{e}_x + r_{y1}\mathbf{e}_y\right)\left(r_{x2}\mathbf{e}_x + r_{y2}\mathbf{e}_y\right)^* \right| \\
&\cos\left[\mathbf{k}\cdot\left(\mathbf{r}_1 - \mathbf{r}_2\right) + \Delta\phi_{12} + \arg\left[\left(r_{x1}\mathbf{e}_x + r_{y1}\mathbf{e}_y\right)\left(r_{x2}\mathbf{e}_x + r_{y2}\mathbf{e}_y\right)^* \right] \right]
\end{aligned}
\tag{16}
$$

Coherence scanning interferometric principle

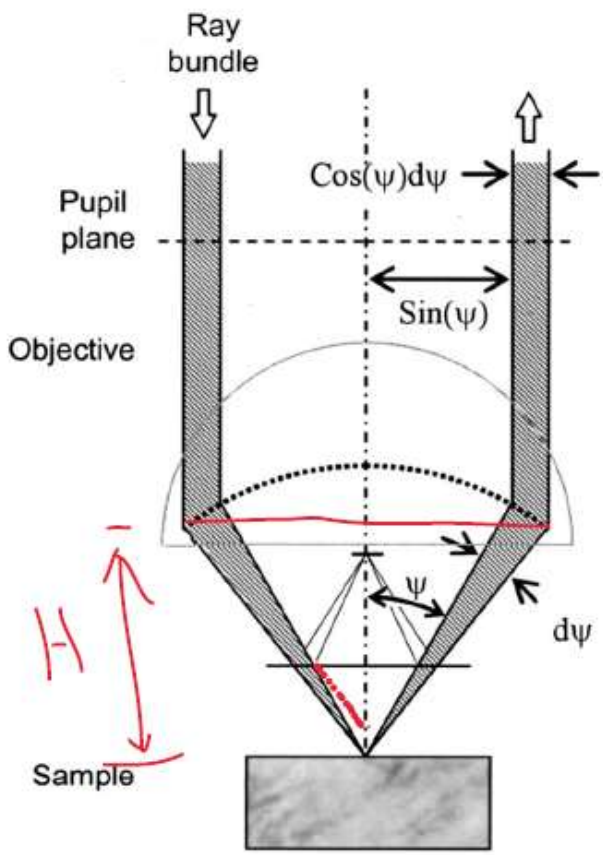


Fig 1 Geometry of a single ray bundle illumination and imaging through an objective obeying the sine condition. The focal length is normalized.

Here，red dashed line denotes the image ray of reflection ray, and the OPD between the reflection ray and the object ray is given as

	$\mathrm{OPD}=\frac{2\pi}{\lambda}2\left(h_s-z\right)\cos\left(\psi\right)=2\pi k\tau\cos\left(\psi\right)$	**(17)**
	$g\left(\beta,k,z,\psi\right)=R_r\left(k,\psi\right)+R_s\left(k,\psi\right)+2\sqrt{R_r\left(k,\psi\right)R_s\left(k,\psi\right)}\cos\left[2\pi k\tau\cos\left(\psi\right)+\Delta\left(k\right)\right]$	**(18)**

where $k=1/\lambda$, $\beta=\cos\left(\psi\right)$, $\tau=2\left(h_s-z\right)$, and $\Delta(k)$ denotes the phase term related with wavenumber. **This phase term could be contributed from the longitudinal chromatic aberration or the complex reflection coefficients from the measured sample.**

Assuming the distance from the pupil plane to the focal plane is denoted as H, and

	$I(z)=\int_0^{\infty}\int_0^{r_0} g(k,z,\psi)U(r)V(k)2\pi r dr dk$	**(19)**
	$r dr=\sin(\psi)\cos(\psi)d\psi$	**(20)**
	$I(z)=\int_0^{\infty}\int_0^{1} g(k,z,\psi)U(\psi)s(k)\sin(\psi)\cos(\psi)d\psi dk$	**(21)**

where $s(k)$ denotes the spectrum distribution of the light source, $U(\beta)$ is the intensity distribution in the pupil plane of the objective. From a physical perspective, the interferometric signal's composition can be understood as the integration over the intensity of the ray bundle at the exit pupil. **This approach is used instead of integrating over the amplitude because we are employing a white light source with low spatial coherence.** **From the expression here, we find there is an implicit assumption, that the intensity distribution at the pupil plane is symmetrical around the optical axis.**

	$I(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}\begin{Bmatrix} R_r(k)+R_s(k)+2\sqrt{R_r(k)R_s(k)} \\ \cos\left[2\pi k\tau\cos(\psi)+\Delta(k)\right]\end{Bmatrix}U(\psi)s(k)\cos(\psi)\sin(\psi)d\psi dk$	**(22)**
	$I(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}\left[R_r(k)+R_s(k)\right]U(\psi)s(k)\cos(\psi)\sin(\psi)d\psi dk$ $+\int_0^{\infty}\int_{\psi_1}^{\psi_2}2\sqrt{R_r(k)R_s(k)}\cos\left[2\pi k\tau\cos(\psi)+\Delta(k)\right]U(\psi)s(k)\cos(\psi)\sin(\psi)d\psi dk$	**(23)**

where $U(\psi)$ denotes the pupil filed distribution for different incidence angle ray, and ψ_1 , and ψ_2 denotes the ray angle limits for a specific interferometric objective.

Uniform pupil field distribution

Assuming we have a uniform distribution among the pupil plane, and the sample reflectance among different incident angles are the same.

	$I(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}\begin{Bmatrix} R_r(k)+R_s(k)+2\sqrt{R_r(k)R_s(k)} \\ \cos\left[2\pi k\tau\cos(\psi)+\Delta(k)\right]\end{Bmatrix}s(k)\cos(\psi)\sin(\psi)d\psi dk$	**(24)**
	$I(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}\left[R_r(k)+R_s(k)\right]s(k)\cos(\psi)\sin(\psi)d\psi dk$ $+\int_0^{\infty}\int_{\psi_1}^{\psi_2}2\sqrt{R_r(k)R_s(k)}\cos\left[2\pi k\tau\cos(\psi)+\Delta(k)\right]s(k)\cos(\psi)\sin(\psi)d\psi dk$	**(25)**

where

	$$I(z)=I_{\text{background}}(z)+I_{\text{fringe}}(z)$$	**(26)**
	$$I_{\text{fringe}}(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}2\sqrt{R_r(k)R_s(k)}\cos\left[2\pi k\tau\cos\psi+\Delta(k)\right]s(k)\cos(\psi)\sin(\psi)d\psi dk$$	**(27)**
	$$I_{\text{background}}(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}\left[R_r(k)+R_s(k)\right]s(k)\cos(\psi)\sin(\psi)d\psi dk$$	**(28)**

CSI model with modulated light source		
	$$I_{\text{fringe}}(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}2\sqrt{R_r(k)R_s(k)}\cos\left[2\pi k\tau\cos\psi+\Delta(k)\right]s(k)M(k,z)\cos(\psi)\sin(\psi)d\psi dk$$	
	$$I_{\text{background}}(z)=\int_0^{\infty}\int_{\psi_1}^{\psi_2}\left[R_r(k)+R_s(k)\right]s(k)M(k,z)\cos(\psi)\sin(\psi)d\psi dk$$	

Matrix simplification

The integration expressions in Eq. (11) and Eq. (12) could be expressed in a discrete form, with a matrix-vector multiplication form.

	$$A(z_i,k_j)=\int_{\psi_1}^{\psi_2}\cos\left[2\pi k_j\tau_i\cos(\psi)+\Delta(k_j)\right]\cos(\psi)\sin(\psi)d\psi$$	**(29)**

where the fringe (without background) measurement matrix could be expressed as

	$$\boldsymbol{A}_{m'n}=\begin{bmatrix} A(z_1,k_1) & A(z_1,k_2) & \cdots & A(z_1,k_{n-1}) & A(z_1,k_n) \\ A(z_2,k_1) & A(z_2,k_2) & \cdots & A(z_2,k_{n-1}) & A(z_2,k_n) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ A(z_{m-1},k_1) & A(z_{m-1},k_2) & \cdots & A(z_{m-1},k_{n-1}) & A(z_{m-1},k_n) \\ A(z_m,k_1) & A(z_m,k_2) & \cdots & A(z_m,k_{n-1}) & A(z_m,k_n) \end{bmatrix}_{m'n}$$	**(30)**

And the fringe (without background) could be expressed as

	$$I_{\text{fringe}}=\boldsymbol{A}\sqrt{R_r(k)R_s(k)}s(k)=\boldsymbol{A}V_1(k)$$	**(31)**

Using the same manner, the background measurement matrix is denoted as

	$$B(z_i,k_j)=\int_{\psi_1}^{\psi_2}\cos(\psi)\sin(\psi)d\psi$$	**(32)**

where

$$\boldsymbol{B}_{m' n} = \begin{bmatrix} B(z_1,k_1) & B(z_1,k_2) & \cdots & B(z_1,k_{n-1}) & B(z_1,k_n) \\ B(z_2,k_1) & B(z_2,k_2) & \cdots & B(z_2,k_{n-1}) & B(z_2,k_n) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ B(z_{m-1},k_1) & B(z_{m-1},k_2) & \cdots & B(z_{m-1},k_{n-1}) & B(z_{m-1},k_n) \\ B(z_m,k_1) & B(z_m,k_2) & \cdots & B(z_m,k_{n-1}) & B(z_m,k_n) \end{bmatrix}_{m' n} \tag{33}$$

And the fringe background can be expressed as

$$I_{\text{background}} = \boldsymbol{B}\left[R_r(k) + R_s(k)\right]s(k) = \boldsymbol{B}V_2(k) \tag{34}$$

Detailed factors in practical applications

unobscured NA

Table1 unobscured NA for different interferometric microscope objective

Magnification	Type	NA	Obscuration
2.5×	Michelson	0.075	0
5×	Michelson	0.13	0
10×	Mirau	0.3	0.09
20×	Mirau	0.4	0.07
50×	Mirau	0.55	0.04
100×	Mirau	0.85	0.27

Non-uniform pupil field distribution

Usually, the pupil filed distribution might be non-uniform, which is not as desirable as the ideal case.

$$A(z_i,k_j) = \int_{\psi_1}^{\psi_2} \cos\left[2\pi k_j \tau_i \cos(\psi) + \Delta(k_j)\right] U(\psi)\cos(\psi)\sin(\psi)\,d\psi \tag{35}$$

It is still evident that the pupil field distribution is symmetrical around the optical axis. In such manner, it is difficult to get an analytic expression of the fringe (without background) measurement matrix. However, it is possible to obtain the (i, j) element in

matrix A using the **numerical integration** technique such as built-in MATLAB function trapz (Trapezoidal numerical integration) at the sacrifice of calculation efficiency. Similar, the background measurement matrix is given as

	$$B\left(z_i,k_j\right)=\int_{\psi_1}^{\psi_2}U\left(\psi\right)\cos\left(\psi\right)\sin\left(\psi\right)d\psi$$	**(36)**

Advanced matrix simplification

Scanning error in practical application

There could be both high frequency and low frequency scanning errors that will result in signal errors. Thus, scanning positions used in practical applications could be expressed as

	$$z_i^{\text{real}} = z_i^{\text{ideal}} + e_i$$	**(37)**

Sample's complex reflection coefficient

The reflection coefficients for non-dielectric materials are complex values and can vary with different incident angles and illumination wavelengths. Thin films represent a unique category and will be discussed in the following section.

	$$\Delta\left(k\right)=\angle r_s=\Delta\varphi\left(k,\psi\right)$$	**(38)**